

MindOS RAG Test Document

This document is created specifically to test the MindOS Knowledge Base, semantic retrieval, embeddings, pgvector, and RAG answer generation.

1. Database Normalization

Database normalization is a process of organizing data in a relational database to reduce unnecessary duplication and improve data integrity. Normalization divides large tables into smaller related tables and establishes relationships between them.

The three commonly discussed normal forms are First Normal Form (1NF), Second Normal Form (2NF), and Third Normal Form (3NF).

2. First Normal Form (1NF)

A table is in First Normal Form when each column contains atomic values and there are no repeating groups. For example, storing multiple phone numbers in one field can violate the atomic-value requirement. The values should instead be represented so that each field contains a single logical value.

3. Second Normal Form (2NF)

A table is in Second Normal Form when it is already in 1NF and every non-key attribute is fully dependent on the entire primary key. 2NF mainly addresses partial dependency, which can occur when a table has a composite primary key.

4. Third Normal Form (3NF)

A table is in Third Normal Form when it is in 2NF and non-key attributes do not depend transitively on the primary key. In simple terms, an attribute should depend on the key, the whole key, and nothing but the key.

5. Benefits of Normalization

- Reduces duplicate data.
- Improves consistency and data integrity.
- Makes database updates safer.
- Reduces update, insertion, and deletion anomalies.
- Creates clearer relationships between related data.

6. Semantic Search Test Facts

MindOS uses semantic retrieval to find document chunks that are conceptually related to a user's question. A question does not always need to use exactly the same words as the document. For example, a question about duplicate records may retrieve the normalization section because reducing duplication is one of normalization's goals.

7. Test Questions

- 1 What is database normalization?
- 2 What are the three normal forms mentioned in this document?
- 3 What is the main requirement of 1NF?
- 4 What problem does 2NF address?
- 5 Explain 3NF in simple terms.
- 6 What are the benefits of normalization?
- 7 Why does normalization help reduce duplicate data?
- 8 What is semantic retrieval according to this document?

Expected RAG Behavior

For questions supported by this document, MindOS should generate an answer grounded in the document and return source metadata such as the document name, chunk information, and similarity score. For a question whose answer is not supported by the knowledge base, the assistant should avoid inventing information and indicate that there is not enough information in the knowledge base.